

Fnu Koustubh

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EDUCATION

Master of Engineering in Robotics

May 2025

University of Maryland - College Park

GPA: 3.8/4.0

Relevant Coursework: Autonomous UAVs, Motion Planning, Computer Vision, Object-Oriented Programming(OOP), Software Development, Control Systems, Reinforcement Learning, Machine Learning and AI

Bachelor of Technology in Electrical Engineering

Aug 2021

Institute of Engineering and Technology, Lucknow

GPA: 8.0/10.0

Relevant Coursework: Embedded Systems, Control Theory, Signals & Systems, Engineering Mathematics

SKILLS

Languages: C++ 14/17, Python 3, MATLAB, CMake

Robotics: ROS/ROS2, PX4 Autopilot, MoveIt, SLAM, Kinematics, Optimization, State Estimation, AutoCAD

Machine Learning and AI: PyTorch, TensorFlow, ScikitLearn, Pandas, Numpy, Transformers, Reinforcement Learning (DQN, PPO, Actor-Critic methods), Point Cloud Library(PCL)

Computer Vision: OpenCV, Vision Language Action models(VLAs), NeRF, Gaussian Splatting, 3D Photogrammetry

Software Tools: Linux, Git, Docker, CI/CD(GitHub Actions, GTest), Jira, Confluence, AWS, GCP, Claude Code

Simulations and Visualization: Gazebo, RViz, FoxGlove, MuJoCo Sim, Isaac Sim, Isaac Lab, CARLA

EXPERIENCE

Robot Operations Engineer | Meta

March 2026 – Present

- Teleoperated robotic systems to collect high-fidelity demonstration data and benchmarked trained VLA model performance on physical robots to ensure reliable real-world execution
- Built and maintained robust teleoperation pipelines, developing real-time KPI dashboards to monitor system performance and data throughput.
- Achieved a 90% system uptime through active software and hardware debugging, successfully increasing the demonstration episode success rate from 80% to 91%.

Software Engineer | Saayam For All

Sept 2025 – March 2026

- Built conversational AI chatbot on AWS Lambda using Flask with Groq's LLaMA 3.1 and Google Gemini fallback, serving 50+ subcategories across social services.
- Engineered RESTful API with automatic category classification, AI-powered subject generation, and conversation history management with robust error handling and CORS support.

Research Assistant | Maryland Applied Graduate Engineering, UMD

Aug 2024 – May 2025

- Mentored two graduate-level robotics courses of 60+ students. Supported the physical installation, configuration, deployment, and troubleshooting of AGV platforms (TurtleBot, Husky).
- Engineered Docker containers to provision one-shot simulation environments, enabling seamless sim-to-real (Sim2Real) software deployments across all hardware.
- Trained users on system operations and verified that all hardware and software punch list items were resolved during the commissioning phase.

Software Engineer | Tata Consultancy Services

Oct 2021 – Aug 2023

- Supported core production systems by optimizing the deployment lifecycle, implementing CI/CD pipelines and version control (Git) to safely push continuous updates.
- Developed internal software tools using Python and Bash scripts to automate diagnostic workflows, successfully reducing Mean Time to Resolution (MTTR) by 65%.

TECHNICAL PROJECTS

UAV-Based 3D Scene Reconstruction | PX4 Autopilot, Docker, AWS, OpenCV

- Optimized a high-performance computer vision pipeline and containerized it on AWS EC2 to accelerate aerial imagery processing from a UAV. Reduced image redundancy by 60% achieving a 15x faster reconstruction.

Autonomous TurtleBot Navigation using Computer Vision | ROS2, OpenCV, C++

- Developed a ROS2 autonomy stack integrating LiDAR, depth cameras, and wrote custom C++ nodes for real-time path planning and obstacle avoidance.

LiDAR SLAM & Pose Graph Optimization | GTSAM, LiDAR, State Estimation, EVO

- Engineered an optimization-based LiDAR SLAM pipeline leveraging GTSAM and loop closure algorithms, successfully reducing Absolute Trajectory Error (ATE) by 38% (3.5m to 2.17m) on the real-world KITTI dataset.